Fundamentals 15

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1. We have that $x \in (a, b)$ iff $x > a$ and $x < b$. Let $x \in (a, b)$ and $\varepsilon = \min(b - x, x - a)$. Now let $y \in \mathbb{R}$ such that $|x - y| < \varepsilon$. If $y \geq x$, we have that

$$|x - y| = y - x < \varepsilon \leq b - x,$$

$$y - x < b - x, \quad y < b.$$

Therefore, $a < x \leq y < b$, so $y \in (a, b)$. If $y < x$, we have that

$$|x - y| = x - y < \varepsilon \leq x - a,$$

$$x - y < x - a, \quad a < y.$$

Therefore, $a < y \leq x < b$, so $y \in (a, b)$.

Now we show that $S = \mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, \infty)$ is open. We know that $x \in S$ iff $x < a$ or $x > b$. Let $x \in S$ and $\varepsilon = \min(a - x, x - b)$. Now let $y \in \mathbb{R}$ such that $|x - y| < \varepsilon$.

First, we consider the case where $x < a$. If $y \leq x$ then $y < a$ so $y \in S$. If $y > x$,

$$|x - y| = y - x < \varepsilon \leq a - x,$$

$$y - x < a - x, \quad y < a,$$

so $y \in S$.

Now, assume $x > b$. If $y \geq x$ then $y > b$ so $y \in S$. If $y < x$,

$$|x - y| = x - y < \varepsilon \leq x - b,$$

$$x - y < x - b, \quad b < y,$$

so $y \in S$.

2. Let $S_n = \left(-\frac{1}{n}, 1 + \frac{1}{n}\right)$, and let $S = \bigcap_{n=1}^{\infty} S_n$. We claim that this is equal to $[0, 1]$.

If $x \in [0,1]$, then $x > -\frac{1}{n}$ and $x < 1 + \frac{1}{n}$ for all $n \in \mathbb{N}$, so any $x \in [0,1]$ is also in S. Now, let $x \in S$, and for contradiction, assume it is not in $[0,1]$. Then, if $x < 0$, we can choose $n > \frac{1}{-x}$, so that $x < -\frac{1}{n}$ and $x \notin S_n$ and therefore $x \notin S$. If $x > 0$, we can choose $n > \frac{1}{x-1}$ so that $x > 1 + \frac{1}{n}$ and $x \notin S$. This is a contradiction, so all elements of S are in $[0,1]$.

3. A finite subset U of the metric space (X,d) consists of points $x_1, x_2, \ldots, x_n$ in order. Let $x_k \in U$. Consider the open ball $B_r(x_k)$ where $r = \min(x_k - x_{k-1}, x_{k+1} - x_k)$. Since any point $x \in X$ is at least r away from x_k, the open ball only contains x_k, meaning it is not a limit point. So U does not contain any limit points.

Also, we have that $\mathbb{R} \setminus U = (-\infty, x_1) \cup (x_1, x_2) \cup \cdots \cup (x_n, \infty)$. Since this is a union of open sets, $\mathbb{R} \setminus U$ is open, so U is closed.

4. Let S be an open neighborhood, and let $x \in S$. Then there is $\varepsilon > 0$ such that for all $y \in X$, $d(x,y) < \varepsilon \implies y \in S$. The set of all y where $d(x,y) < \varepsilon$ is the open ball $B_\varepsilon(x)$, and we have that $y \in S$ for all such y, so $B_\varepsilon(x) \subseteq S$.

5. Since x is a limit point of U, for all $\varepsilon > 0$, the open ball $B_\varepsilon(x)$ contains at least one point of U different from x. We construct a sequence (x_n) in U that converges to x:
For each $n \in \mathbb{N}$, there exists a point $x_n \in U$ such that $x_n \neq x$ and $d(x, x_n) < \varepsilon_n$.

To show that this converges to x, let $\varepsilon > 0$ be arbitrary, and let $N > \frac{1}{\varepsilon}$. Then for all $n > N$,

$$d(x, x_n) < \frac{1}{n} < \frac{1}{N} < \varepsilon,$$

so (x_n) converges to x.

6. Assume $S \subseteq \mathbb{R}$ is open and closed. Then S and S^c are open. Take any element $x \in S$ (unless $S = \varnothing$). Since S is open, there exists $\varepsilon > 0$ such that for all $y \in \mathbb{R}$, $|x - y| < \varepsilon \implies y \in S$.

First, we consider values for y that are $> x$. Let $B = \{\varepsilon : y > x, |x - y| < \varepsilon \implies y \in S\}$. Since S is open, there is an open ball centered at $x + \varepsilon$ completely contained in S. So there always exists $\varepsilon' > \varepsilon$ that works as well.

Let $a = \sup(B)$ (unless B is unbounded above, meaning all neighborhoods containing x are unbounded above) and $x' = x + a$. Since B has no greatest element, $x' \notin S$, so $x' \in S^c$. Since S^c is closed, there exists $\varepsilon > 0$ such that for all $y \in \mathbb{R}$, if $|x' - y| < \varepsilon$ then $y \in S^c$.

Choose $\varepsilon > 0$ so that it satisfies the above, and also so that $x' - \varepsilon \in S$. This exists because $x' = x + a$ where a is the least possible number such that

$x + a \in S^c$. Then let y be such that $|x' - y| < \varepsilon$, and therefore, $y \in S^c$. But this is a contradiction.

By a similar argument where $y < x$, S cannot be open and closed unless all neighborhoods containing x are unbounded below. So we have that S is open and closed only if $S = \varnothing$ or $S = \mathbb{R}$.

8. Let $\mathcal{O}$ be an open cover of $S = \{1/n : n \in \mathbb{N}\} \cup \{0\}$. We know that $\mathcal{O} = \{U_\alpha : \alpha \in A\}$ for some set A and open sets U_α. We need to find $F \subseteq A$ where F is finite so that $\{U_\alpha : \alpha \in F\}$ also covers S.

Let $\alpha \in A$ be such that U_α contains 0 and let $\alpha \in F$. Since U_α is open, it includes all x where $|x| < \varepsilon$ for some ε. So for all $n > \frac{1}{\varepsilon}$, $\frac{1}{n} \in U_\alpha$. The only remaining terms in S to cover are $\frac{1}{n}$ where $1 \leq n \leq \frac{1}{\varepsilon}$, which are finitely many. Constructing F this way involves putting finitely many terms, so $\{U_\alpha : \alpha \in F\}$ is a finite subcover of $\mathcal{O}$.

9a. All subsets U of (X, d) are open because, for all $x \in U$, we can set $\varepsilon = \frac{1}{2}$ so that if $d(x, y) < \varepsilon$ then $x = y$, which means $y \in U$.

9b. Consider a finite subset $U \subseteq X$. Any open cover of U consists of open sets that include each element of U. Since U is finite, we can select a finite number of these open sets to cover U.

If $U \subseteq X$ is infinite, consider the open cover given by $\{\{x\} : x \in U\}$. This is an open cover since $\{x\}$ is open. But no finite collection of these sets can cover U (since each set in the open cover contains exactly one point and U is infinite), so U is not compact. Therefore, U is compact iff it is finite.

9c. A function f is continuous if for all $x \in X$ if for all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon.$$

We can always set $\delta = \frac{1}{2}$, so that $d(x, y) < \delta \implies x = y$, so $d(f(x), f(y)) = 0 < \varepsilon$ regardless of ε. So every function is continuous under the discrete metric.

12. $f^{-1}(f(U)) = U$ when f is surjective, and $f(f^{-1}(V)) = V$ when $V \subseteq \{f(x) : x \in \mathbb{R}\}$. For this to be true for all V, the range of f must be $\mathbb{R}$.